

Languages, automata, and the Chomsky hierarchy, continued

Computational modeling of language
Spring 2026

Bureaucracy

Bureaucracy

- **Waitlist:** All done!

Bureaucracy

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- **In case of illness:** Please stay home and get better - we will accommodate!

Last week

- FSAs and regular languages
- The Chomsky hierarchy
- The halting problem and theoretical limits to computability
- Relevance to human language

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This week

- The pumping lemma
- PDAs and context-free languages
- Chomsky vs. Shannon on human language
- On toward numbers

Small groups

Trade notes on **last week's material**,
and **this week's readings**, to generate
comments and/or questions.

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- Chomsky vs. Shannon on human language
- On toward numbers

Overview

- Beyond regular languages
- Context-free languages and pushdown automata
- The argument of Chomsky 1956
- On toward numbers

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baa!
baaa!
baaaa!
baaaaa!
...

$S \rightarrow bT$
 $T \rightarrow aU$
 $U \rightarrow aV$
 $V \rightarrow aV \mid !$

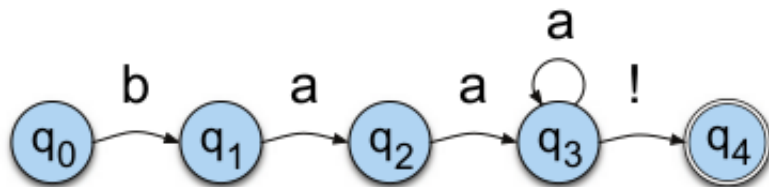
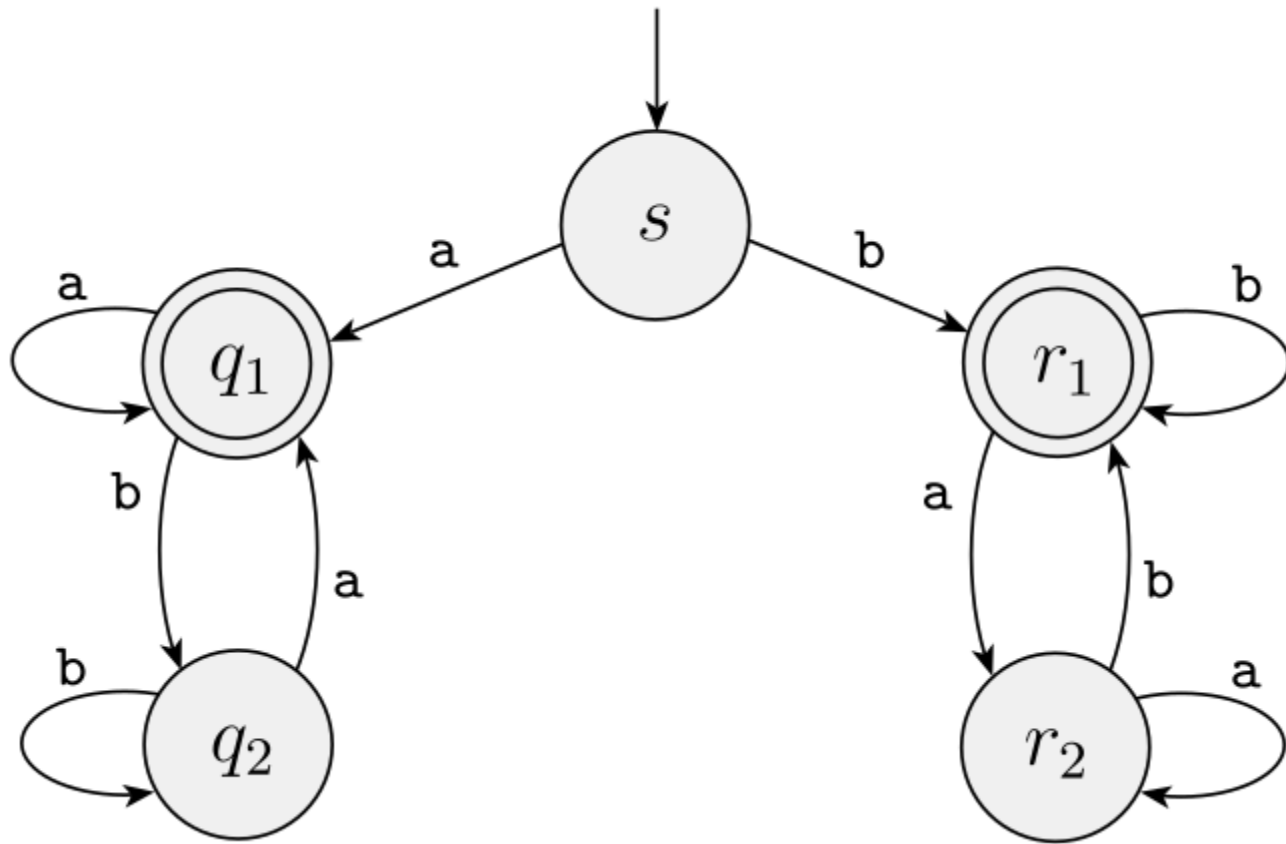


Figure 2.10 A finite-state automaton for talking sheep.



Example: SPE-style rules in phonology

General form:

$$A \rightarrow B \quad / \quad C __ D$$

Simple finite-state **transducer**:



What is **not** a regular language?

And how can you tell?

Pumping lemma for regular languages

Pumping lemma for regular languages

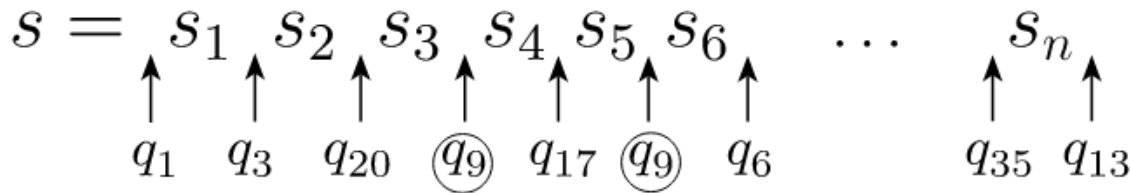
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Pumping lemma for regular languages

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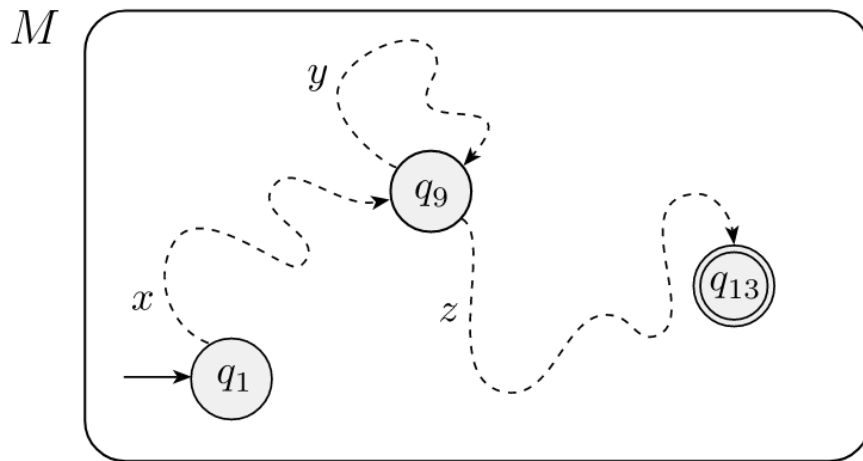
Pumping lemma for regular languages

- Let L be a regular language. Then it is recognized by some FSM M , with n states.
- Consider a string $s \in L$ of length $\geq n$.
- s must cause M to **revisit** at least one state - i.e. there must be a **loop** as we trace s through M .



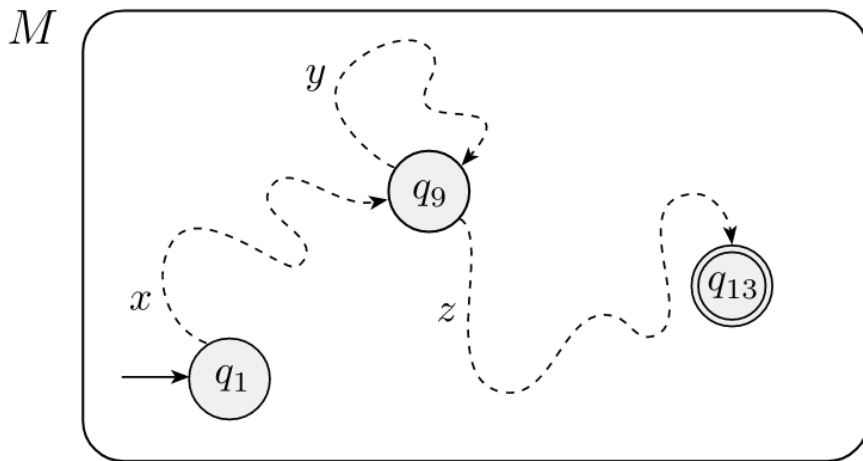
Pumping lemma for regular languages

- Therefore $s = xyz$:



Pumping lemma for regular languages

- Therefore $s = xyz$:



- And therefore $x(y^i)z \in L, \forall i \geq 0$.

Consider the simple language $\{0^n 1^n \mid n \geq 0\}$

- Any number of 0s followed by the same number of 1s.

Consider the simple language $\{0^n 1^n \mid n \geq 0\}$

- Any number of 0s followed by the same number of 1s.
- For example:
 - 01
 - 0011
 - 000111
 - 00001111
 - ...

$\{0^n 1^n \mid n \geq 0\}$ is not regular

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Proof by contradiction

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 $s = xyz$, s.t. $x(y^i)z$ is also in this language.

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What we've seen so far

- The pumping lemma.
- How to use the pumping lemma to prove that certain languages are not regular.
- A particular non-regular language, $\{0^n 1^n \mid n \geq 0\}$.

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- How to use the pumping lemma to prove that certain languages are not regular.
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Often expressed as $\{\mathbf{a^n b^n} \mid n \geq 0\}$

A question you might have

- Regular grammars have rules of the form:
 - $A \rightarrow a$
 - $A \rightarrow \varepsilon$
 - Either:
 - $A \rightarrow aB$ (right regular), or
 - $A \rightarrow Ba$ (left regular)

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- Regular grammars have rules of the form:
 - $A \rightarrow a$
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 - Either:
 - $A \rightarrow aB$ (right regular), or
 - $A \rightarrow Ba$ (left regular)
- Q: What if we **mix** right-regular and left-regular rules in a single grammar? A: **Not** regular!

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- Consider this grammar, which mixes left- and right-regular rules:
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 - $A \rightarrow Sb$
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A question you might have

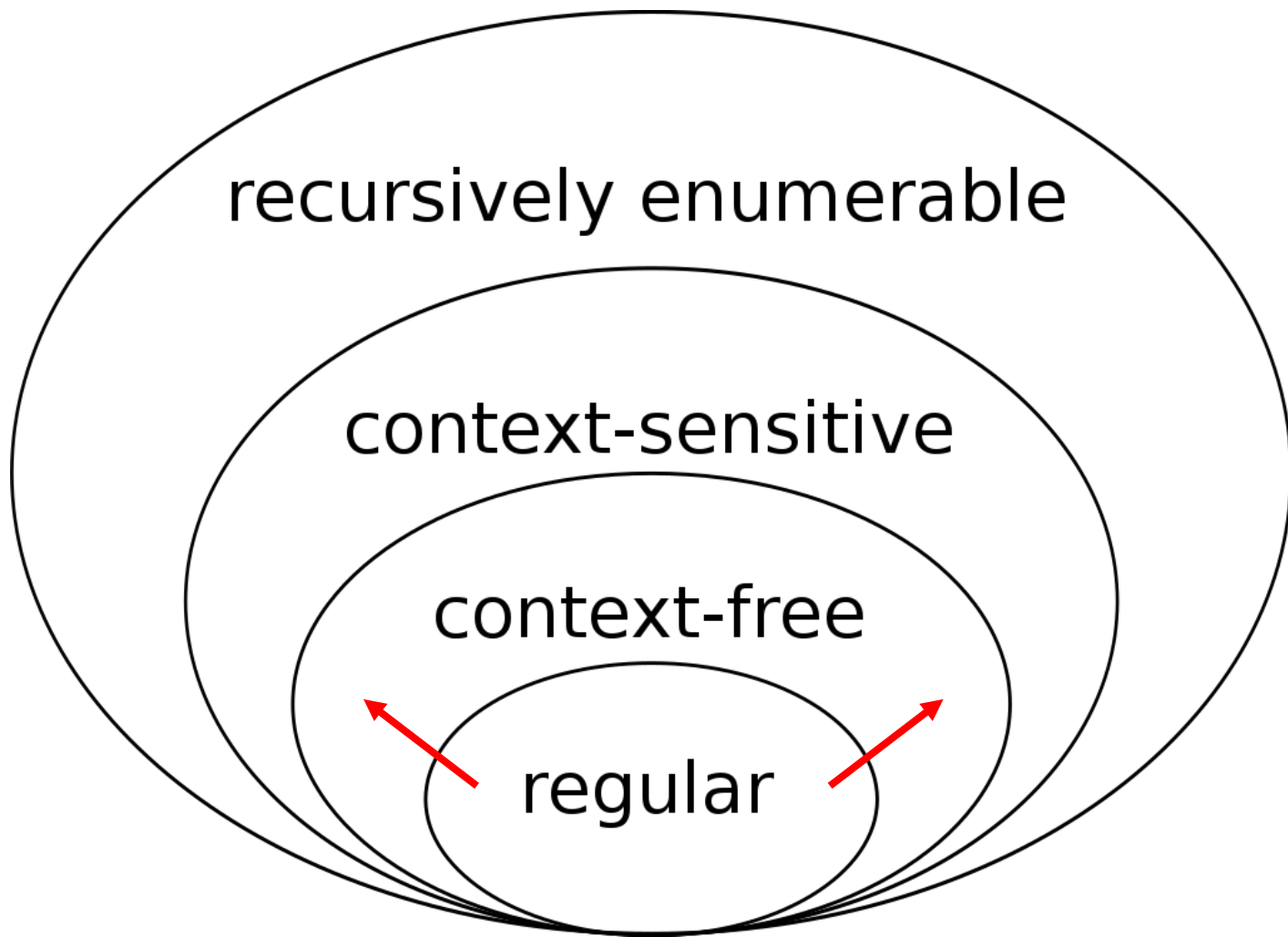
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- Consider this grammar, which mixes left- and right-regular rules:
 - $S \rightarrow aA$
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- What language does this grammar generate?
 - $\{a^n b^n \mid n \geq 0\}$, which is now our canonical **non-regular** language!

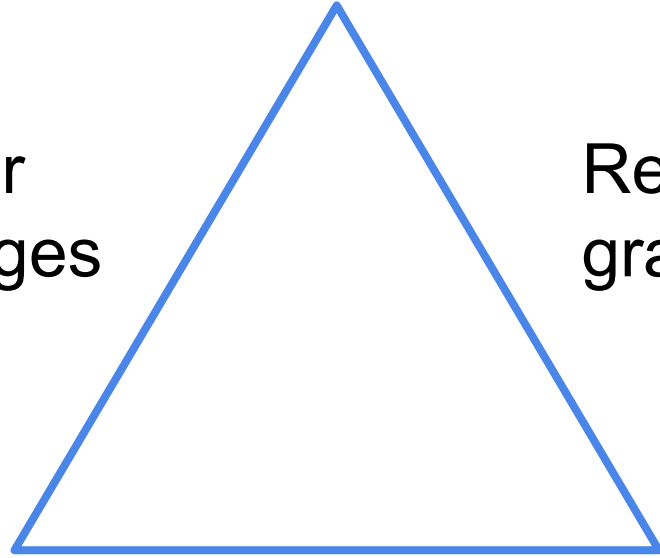
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- **Context-free languages and pushdown automata**
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Regular
languages

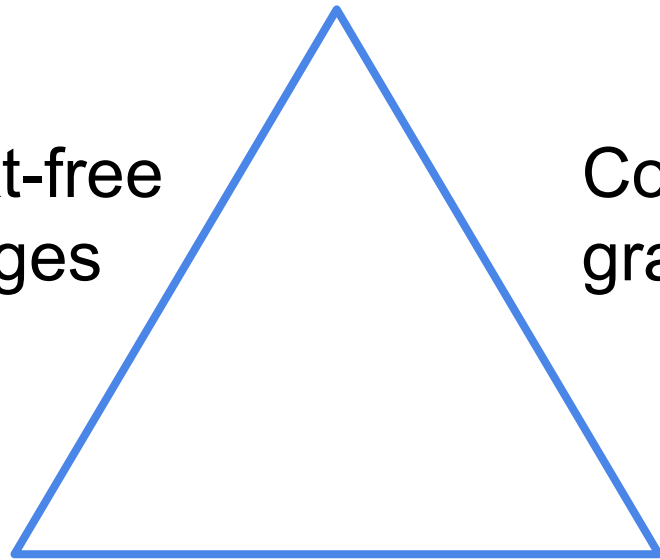
Regular
grammars



Finite state
automata

Context-free
languages

Context-free
grammars



Pushdown
automata

A simple context-free grammar

$$A \rightarrow 0A1$$

$$A \rightarrow B$$

$$B \rightarrow \#$$

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A derivation:

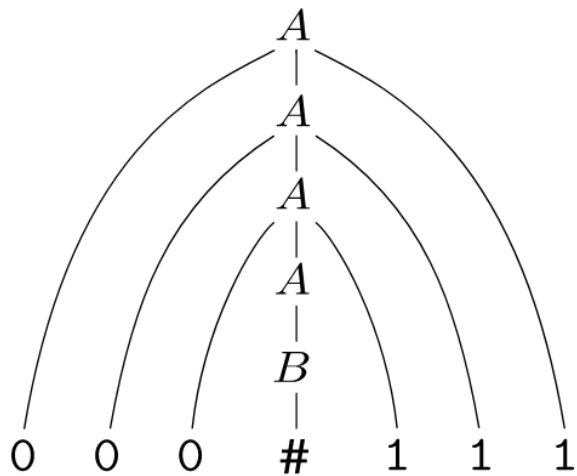
$$A \Rightarrow 0A1 \Rightarrow 00A11 \Rightarrow 000A111 \Rightarrow 000B111 \Rightarrow 000\#111.$$

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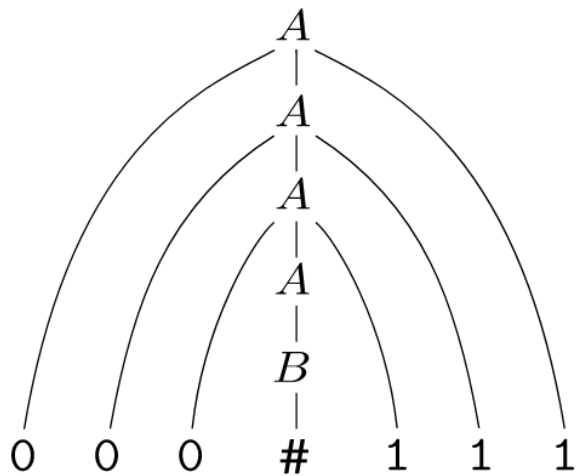
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**Center
embedding**



A derivation:

$$A \Rightarrow 0A1 \Rightarrow 00A11 \Rightarrow 000A111 \Rightarrow 000B111 \Rightarrow 000\#111.$$

Another context-free grammar

$\langle \text{SENTENCE} \rangle \rightarrow \langle \text{NOUN-PHRASE} \rangle \langle \text{VERB-PHRASE} \rangle$
 $\langle \text{NOUN-PHRASE} \rangle \rightarrow \langle \text{CMPLX-NOUN} \rangle \mid \langle \text{CMPLX-NOUN} \rangle \langle \text{PREP-PHRASE} \rangle$
 $\langle \text{VERB-PHRASE} \rangle \rightarrow \langle \text{CMPLX-VERB} \rangle \mid \langle \text{CMPLX-VERB} \rangle \langle \text{PREP-PHRASE} \rangle$
 $\langle \text{PREP-PHRASE} \rangle \rightarrow \langle \text{PREP} \rangle \langle \text{CMPLX-NOUN} \rangle$
 $\langle \text{CMPLX-NOUN} \rangle \rightarrow \langle \text{ARTICLE} \rangle \langle \text{NOUN} \rangle$
 $\langle \text{CMPLX-VERB} \rangle \rightarrow \langle \text{VERB} \rangle \mid \langle \text{VERB} \rangle \langle \text{NOUN-PHRASE} \rangle$
 $\langle \text{ARTICLE} \rangle \rightarrow \text{a} \mid \text{the}$
 $\langle \text{NOUN} \rangle \rightarrow \text{boy} \mid \text{girl} \mid \text{flower}$
 $\langle \text{VERB} \rangle \rightarrow \text{touches} \mid \text{likes} \mid \text{sees}$
 $\langle \text{PREP} \rangle \rightarrow \text{with}$

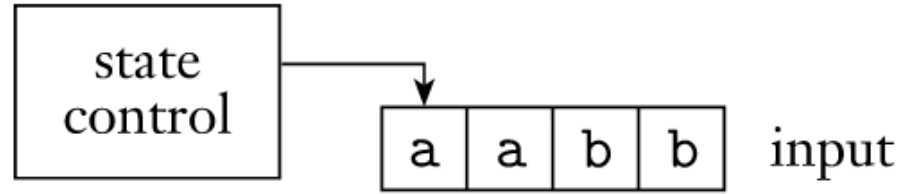
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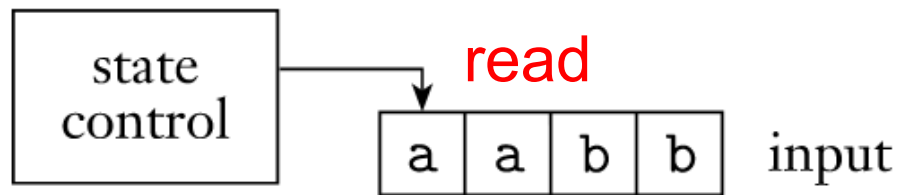
A derivation:

$\langle \text{SENTENCE} \rangle \Rightarrow \langle \text{NOUN-PHRASE} \rangle \langle \text{VERB-PHRASE} \rangle$
 $\Rightarrow \langle \text{CMPLX-NOUN} \rangle \langle \text{VERB-PHRASE} \rangle$
 $\Rightarrow \langle \text{ARTICLE} \rangle \langle \text{NOUN} \rangle \langle \text{VERB-PHRASE} \rangle$
 $\Rightarrow \text{a} \langle \text{NOUN} \rangle \langle \text{VERB-PHRASE} \rangle$
 $\Rightarrow \text{a boy} \langle \text{VERB-PHRASE} \rangle$
 $\Rightarrow \text{a boy} \langle \text{CMPLX-VERB} \rangle$
 $\Rightarrow \text{a boy} \langle \text{VERB} \rangle$
 $\Rightarrow \text{a boy sees}$

A finite state automaton:

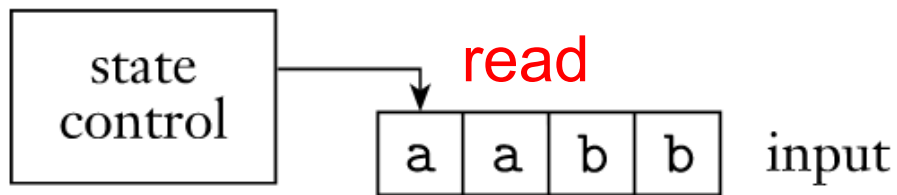


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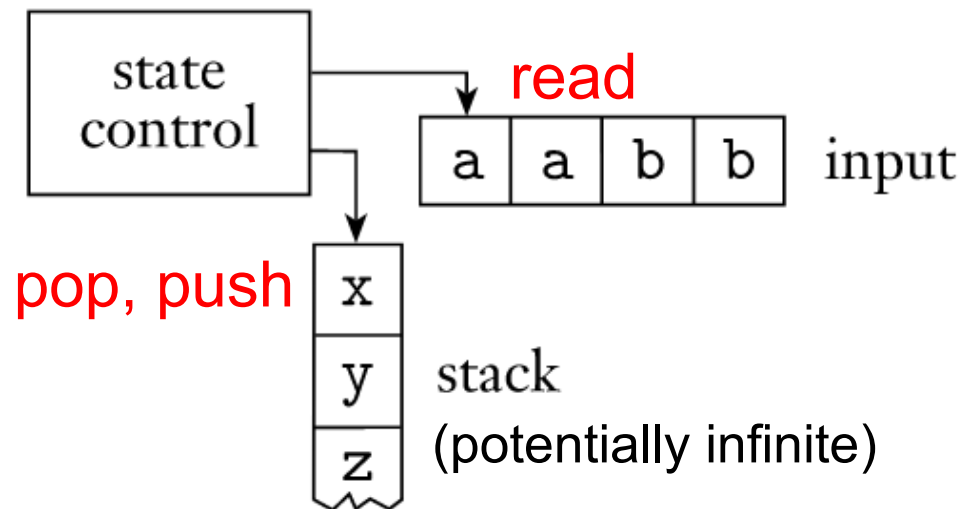


Basic operations

A finite state automaton:

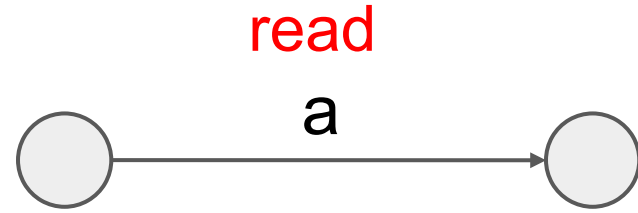


A pushdown automaton:

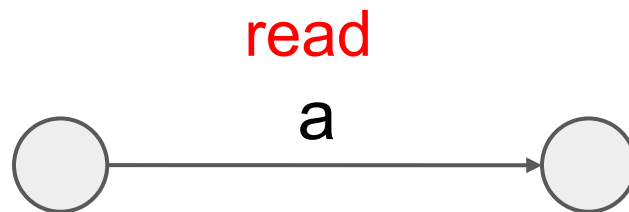


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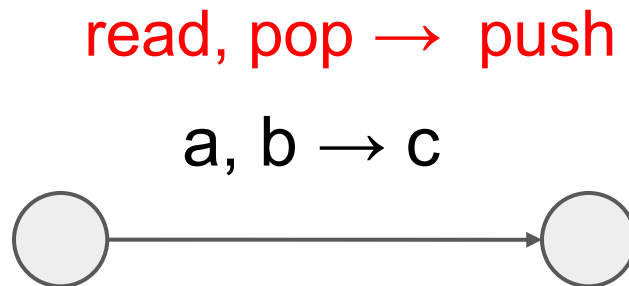
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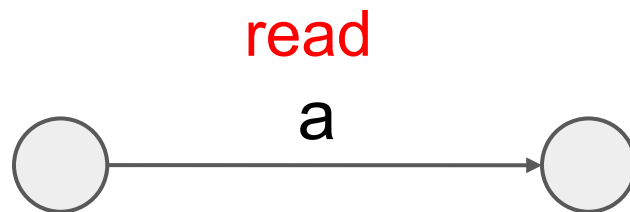
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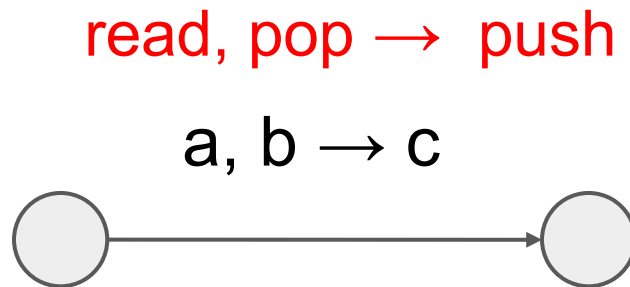
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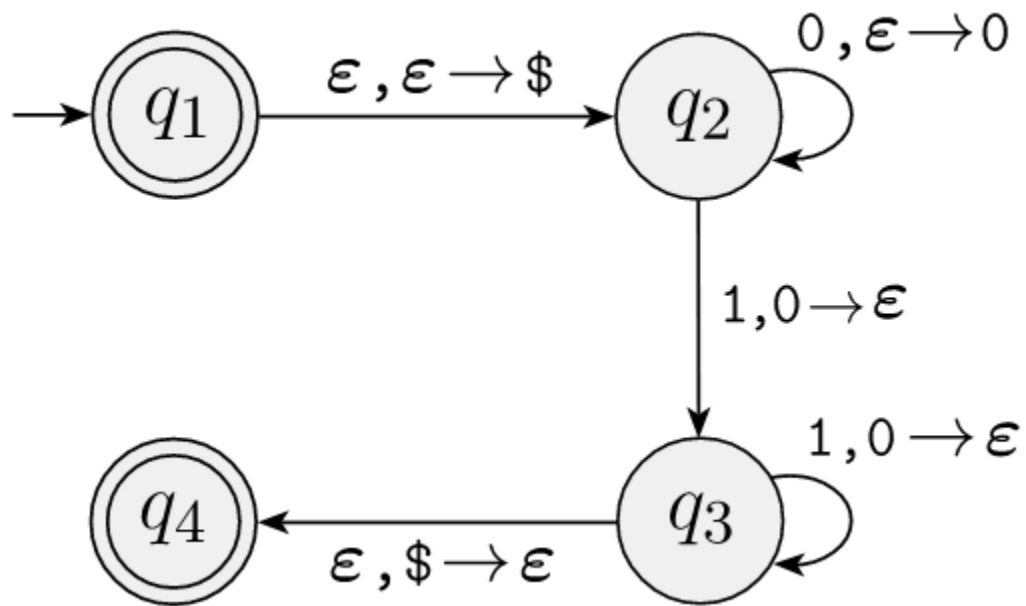
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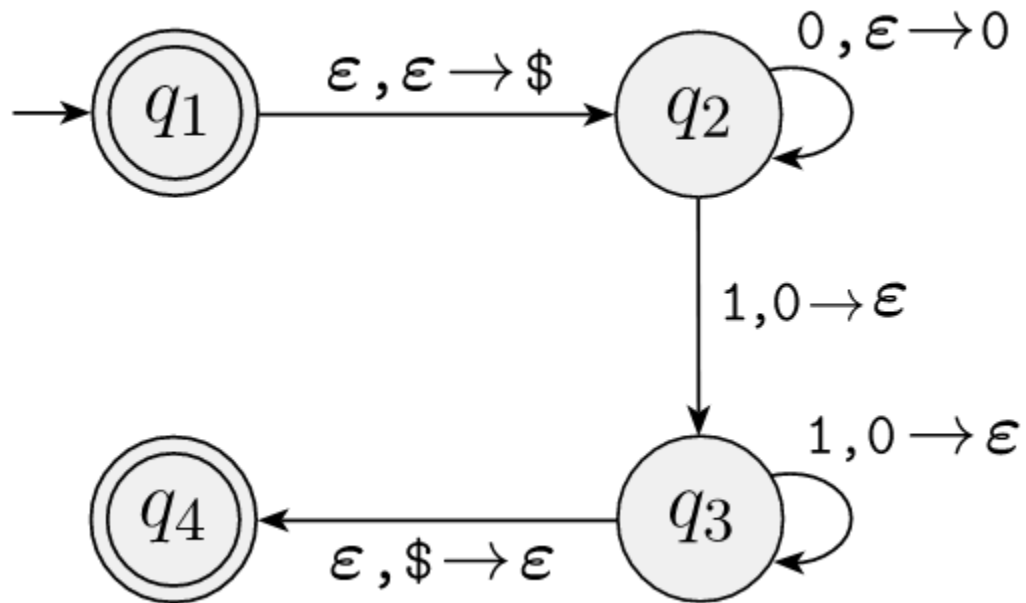
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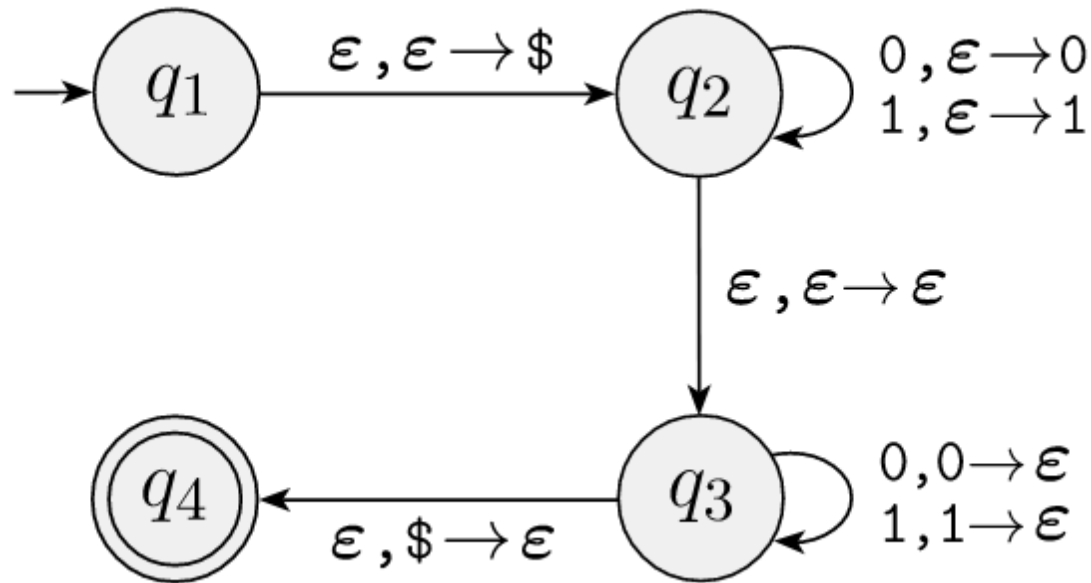


Nondeterministic!

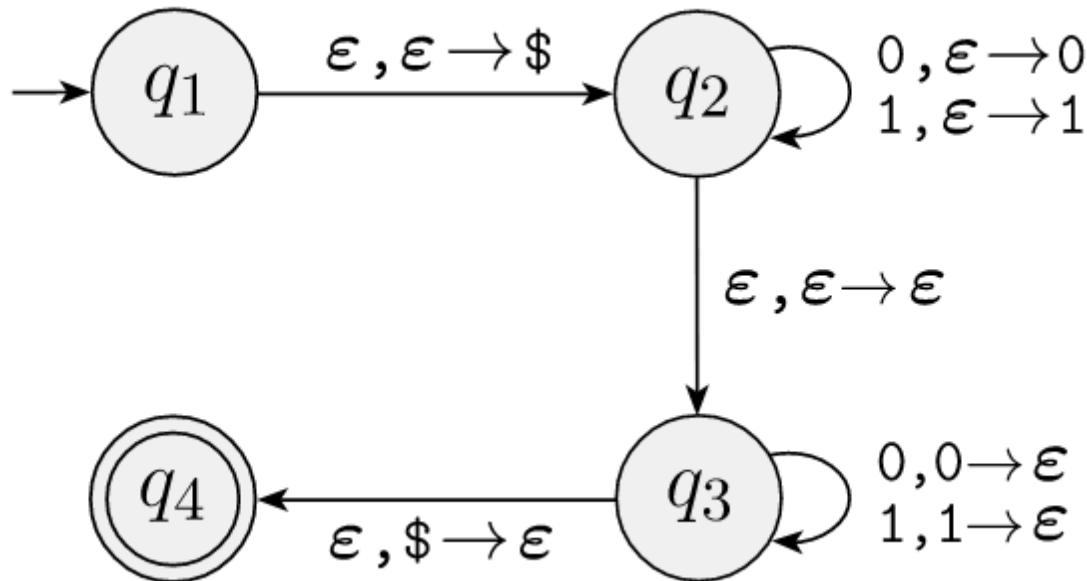


PDA that recognizes $\{0^n 1^n \mid n \geq 0\}$





PDA that recognizes $\{ww^R \mid w \in \{0,1\}^*\}$



Wait! I'm a linguist! What does this have to do with human language?

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Three models for the description of language (really, of English syntax)

1. Finite state machines / Markov chains
2. Phrase structure grammar
3. Transformational grammar

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Three models for the description of language (really, of English syntax)



1. Finite state machines / Markov chains



2. Phrase structure grammar: declaratives

3. Transformational grammar: everything else

Why did Chomsky consider FSMs?

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Response to **Shannon and Weaver 1949**: *The mathematical theory of communication*.

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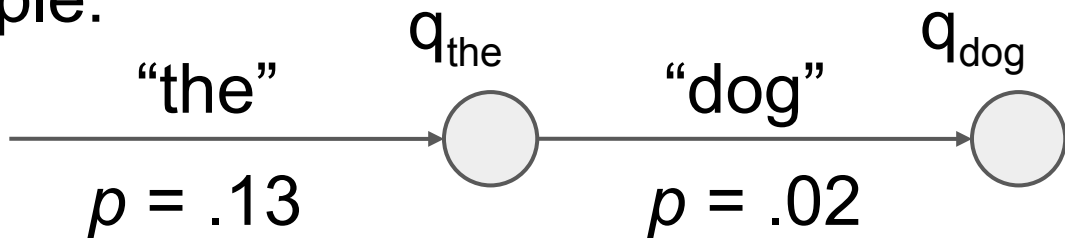
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Cf. Shannon 1948: A mathematical theory of communication.

Why did Chomsky consider FSMs?

Response to **Shannon and Weaver 1949**: *The mathematical theory of communication*.

- S&W considered successive statistical approximations to English, produced by simple unigram and bigram models, trained on English text, for example:



Unigram model sample output:

- REPRESENTING AND SPEEDILY IS AN GOOD APT
OR COME CAN DIFFERENT NATURAL HERE HE
THE A IN CAME THE TO OF TO EXPERT GRAY
COME TO FURNISHES THE LINE MESSAGE HAD
BE THESE.

Bigram model sample output:

- THE HEAD AND IN FRONTAL ATTACK ON AN
ENGLISH WRITER THAT THE CHARACTER OF
THIS POINT IS THEREFORE ANOTHER METHOD
FOR THE LETTERS THAT THE TIME OF WHO
EVER TOLD THE PROBLEM FOR AN
UNEXPECTED.

Shannon and Weaver 1949:

- ‘The resemblance to ordinary English text increases quite noticeably ... The particular sequence of ten words “attack on an English writer that the character of this” is not at all unreasonable.’ (p. 44)

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- ‘The resemblance to ordinary English text increases quite noticeably ... The particular sequence of ten words “attack on an English writer that the character of this” is not at all unreasonable.’ (p. 44)
- Thus, S&W implicitly suggested that higher-order statistical approximations may eventually converge to “ordinary English text”.

Why are FSMs inadequate for English syntax?

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Consider Chomsky's example 13(iii):

- The man who said that S5 is arriving today.

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Now let S5 be:

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Yielding:

- ... man ... women ... S6 ... are ... is ...

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- ... man ... women ... S6 ... are ... is ...

Center
embedding!

Major impacts of Chomsky 1956

1. English syntax is not regular, and requires at least **phrase structure** (e.g. CFGs).

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1. English syntax is not regular, and requires at least **phrase structure** (e.g. CFGs).
2. More generally, redirected attention toward structure, and **away from statistics**.

Colorless green ideas sleep furiously.

Google Books Ngram Viewer

colorless green ideas sleep furiously

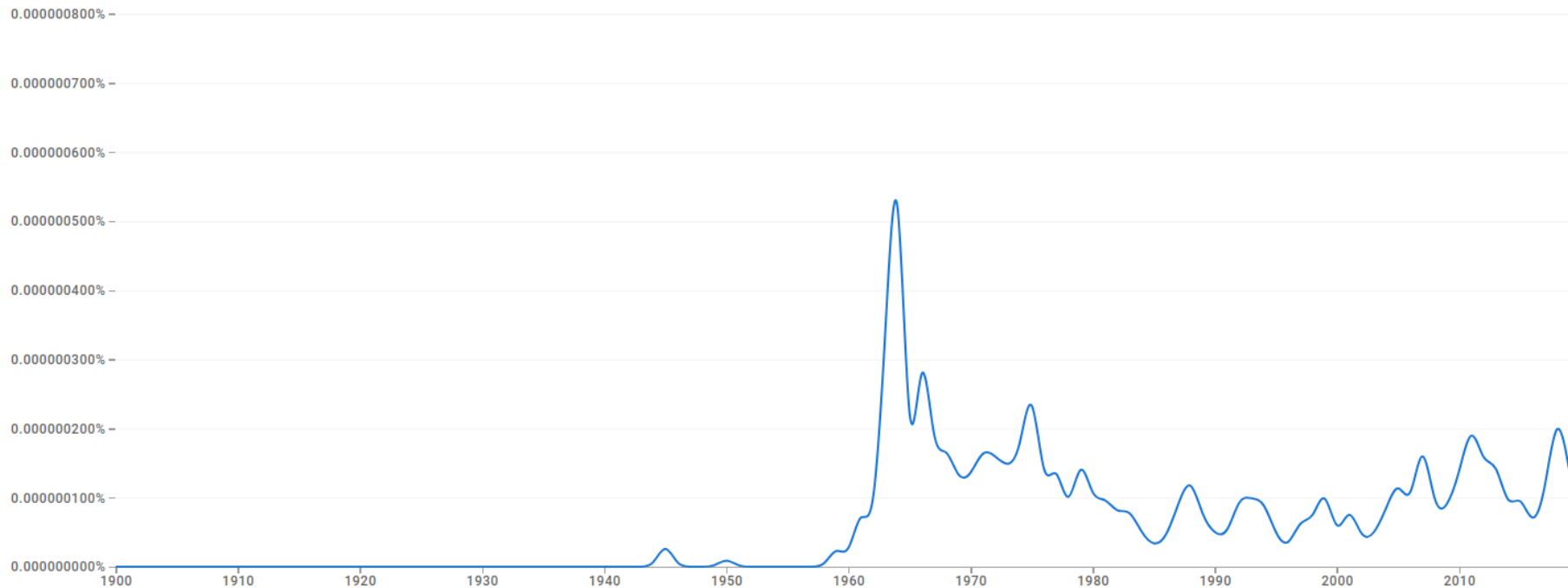


1900 - 2019

English (2019)

Case-Insensitive

Smoothing of 0



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- Suggests to Chomsky that:
 - “the notion of statistical approximation appears to be irrelevant to grammar.”

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- Suggests to Chomsky that:
 - “the notion of statistical approximation appears to be irrelevant to grammar.”

Is this claim **warranted**? **Why** or **why not**?

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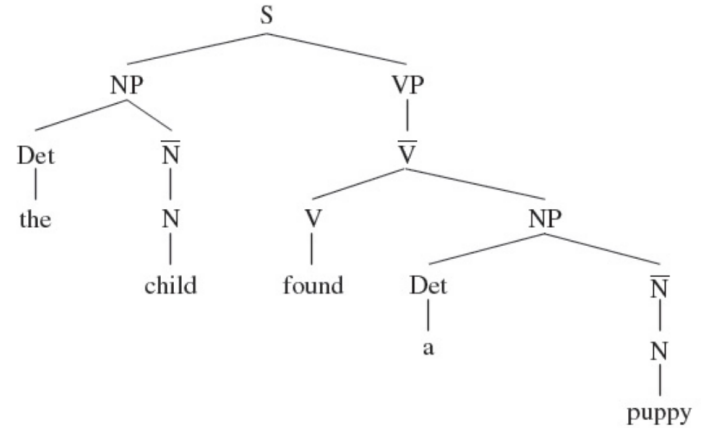
Possible relations of structure and numbers

Arranged in a cline of increasing importance for numbers

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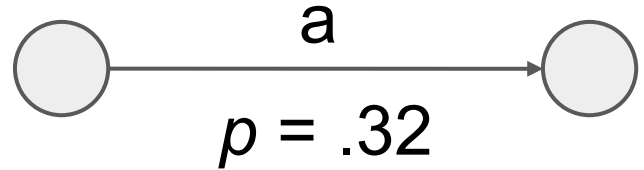
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Possible relations of structure and numbers

Arranged in a cline of increasing importance for numbers

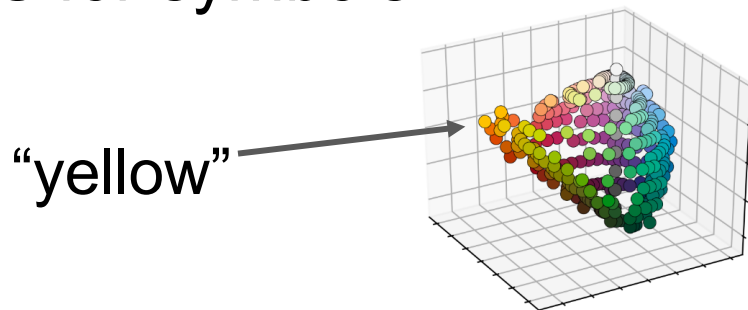
- All symbols, **no** numbers
- Symbols **augmented** by numbers



Possible relations of structure and numbers

Arranged in a cline of increasing importance for numbers

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